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My Learning Experience

I completed this data project. I want to show my work and my thoughts. I did 100% of this work alone.

Initial Setup and Split

First, I chose to use one GPU, not CPU, to work fast. I downloaded this raw Ames Housing data from Kaggle. I uploaded this CSV file to my Google Colab. I never opened or changed this file in Excel. Every change was done using Python code. I split this data into training (70%) and testing (30%). I looked at this result. I saw over 2000 rows in this training set. This size is good. I decided to ignore this test data completely. This test data does not exist for this model right now. This choice stops data leakage.

Cleaning and Imputation

I searched for missing values. I changed 1904 NaN values to 'None' for this Alley feature. I looked at this result to confirm this change. I filled missing numbers for this Lot Frontage feature. I chose to use median numbers, not average numbers. Median numbers are better because they ignore very high outliers.

Feature Analysis and Growth

I transformed text into numbers. I wanted to see all feature connections. I built one big correlation map. I looked at this map and saw too much data. It was hard to read. I decided to make one smaller map to show only connections with SalePrice. This second map is much better.

Normalization, Scaling and Balancing

I balanced these target classes. I plotted two charts for this target balance. I looked at these charts and decided I needed to see more information. I added one third chart to show this SalePrice distribution. This final result shows my complete thought process.

ML Problem

I included one discussion block about using this clean dataset to solve one machine learning problem for real estate.

Team Contribution

I worked alone on this project. I did 100% of this work.